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Language-Guided Manifold Learning for Cross-Domain Remote Sensing Object Detection

Haoxiang Chen, Wei Zhao, Xiping Shang, Rufei Zhang, Nannan Li, and Dongjin Li

***Abstract*—Cross-domain remote sensing object detection under the Single-Domain Generalized Object Detection (SDGOD) setting is crucial for deploying detectors in unseen target domains. Existing methods typically perform feature fusion and matching in Euclidean space with a fixed inner-product metric, which may overlook feature geometry and cause biased similarity estimation under domain shift. Even when structured geometry is modeled, maintaining manifold feasibility across layers remains challenging. Moreover, instance-level text in detection and localization scenarios is often too short and sparse to provide reliable cross-domain semantic anchors. To address the above issues, we propose a Language-Guided Manifold Learning Network (LMLNet), a Vision-Language Model (VLM) framework for cross-domain generalizable object detection in remote sensing. First, we propose Affine-Invariant Riemannian Metric Calibrated Cross-Attention (AccA), which models feature geometry with an AIRM-induced metric to reduce the similarity-estimation bias caused by fixed inner-product matching under cross-domain shifts. Second, we propose a Geometry-Preserving Bilinear Mapping (GPB) with Stiefel-constrained optimization to preserve geometric constraints during network forward propagation. Finally, we propose a Text Denoising Key-Value (TDKV) mechanism, which introduces textual attribute noise to enhance the robustness of text guidance under weak instance-level semantics. Experiments on M^3VL and DIOR2HRRSD datasets demonstrate the effectiveness and cross-dataset generalizability of LMLNet. The project resources are available at https://github.com/softwarePupil/LMLNet**

***Index Terms*—Single-Domain Generalized Object Detection (SDGOD), Manifold Learning, Vision-Language Model (VLM), Affine-Invariant Riemannian Metric (AIRM), remote sensing.**

I. Introduction

CROSS-DOMAIN remote sensing object detection aims to maintain robust detection performance when the training and deployment domains differ. In the single-domain setting, Single-Domain Generalized Object Detection (SDGOD) trains detectors on only one source domain and requires them to generalize to inaccessible target domains [1]. As shown in Fig. 1(a), diverse imaging conditions and scene variations introduce complex domain shifts, making generalization highly challenging.

Recent domain-generalizable object detection methods have explored diverse strategies to improve robustness to unseen domains [2]–[6]. Among them, augmentation and consistency-based methods expand source-domain variations and stabilize training through feature alignment or consistency constraints [2], [3]. Other methods enhance robustness through debiasing, causal reasoning, self-supervision, or generative modeling [4], [5].

Vision-language methods [6]–[10] are promising for SDGOD because a pre-trained vision-language model (VLM) learns transferable image-text alignment from large-scale image-text pairs. Since target-domain images are inaccessible during training, while textual descriptions and semantic attributes are easier to obtain, text can provide complementary cues to improve transferability toward unseen domains [6]–[8]. However, existing Vision-Language Models still lack geometric awareness and stable cross-domain semantic guidance, leading to three key challenges:

1) Metric bias is induced by Euclidean cross-modal matching. Feature representations often exhibit geometric structures shaped by feature correlations [11]. In VLM-based detectors, these visual structures need to be matched with textual semantics during cross-modal fusion, where similarity estimation directly affects classification and localization decisions. However, existing fusion heads usually compute visual-textual similarity in Euclidean space with a fixed inner-product metric. As shown in Fig. 1(b), this metric is unsuitable for geometrically structured features. In cross-domain scenarios, as illustrated in Fig. 1(c), the local affine perturbation (LAP) model introduced in Section III-B indicates that domain shifts can locally transform feature distributions. Such perturbations can bias Euclidean metric-based similarity estimation and degrade detection performance.

2) Geometric constraints are difficult to preserve in deep networks. Explicit feature-geometry modeling requires structured representations to satisfy their geometric feasibility constraints [13]. However, layer-wise mappings, normalization, and numerical errors may drive intermediate representations away from the feasible set, causing geometric degeneration during network forward propagation.

3) Instance-level text provides insufficient semantic guidance. Instance-level prompts in object detection are often short phrases or category names with limited information and weak discriminability [14]. When object appearance changes in target domains, such sparse text fails to provide reliable semantic anchors, which weakens cross-domain text guidance.

To address the above challenges, we propose a Language-Guided Manifold Learning Network (LMLNet), which consists of three key modules: the Affine-Invariant Riemannian Metric (AIRM)-Calibrated Cross-Attention (AccA), the Geometry-Preserving Bilinear Mapping (GPB), and the Text Denoising Key-Value (TDKV).

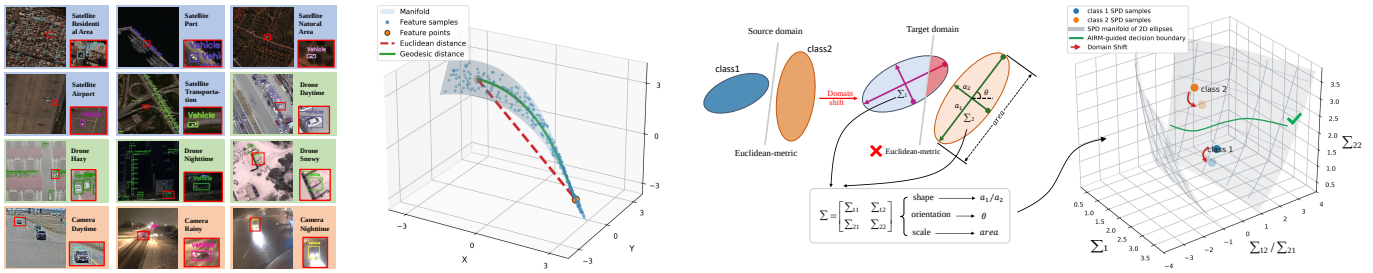
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(a) (b) (c) (d)

Fig. 1. Importance of manifold geometry in feature modeling. (a) Representative vehicle samples from 12 scenarios with platform, weather, and scene variations. (b) Euclidean distance may cross sample-free regions, whereas geodesic distance better follows the manifold structure. For clarity, (c) and (d) use 2D covariance ellipses and their SPD representations for illustration. (c) Under domain shifts, Euclidean decision boundaries may fail under distorted feature distributions. Second-order moments encode shape, orientation, and scale, providing geometry-aware cues under domain shifts. (d) SPD representations are visualized by Σ_{11}, Σ_{12}, and Σ_{22}, where red arrows denote domain shift and the green curve indicates the AIRM-guided decision boundary.

The AccA module is designed to overcome the fixed-metric limitation of Euclidean image-text matching. It represents image and text features with second-order covariance statistics and constructs a shared symmetric positive definite (SPD) metric state on the SPD manifold. As illustrated in Fig. 1(d), covariance descriptors encode the shape, orientation, and scale of feature distributions, while AIRM provides an affine-invariant geometry for comparing such SPD structures [15]. Therefore, AccA replaces the Euclidean inner product with an AIRM-guided inner product, enabling image-text fusion to maintain more stable similarity measurement under cross-domain perturbations.

The GPB module is designed to maintain the feasibility of SPD metrics. In forward propagation, GPB preserves the positive-definite structure of SPD matrices through a full-rank bilinear congruence mapping. In backward updates, Stiefel-constrained optimization keeps the mapping matrix semi-orthogonal, preventing rank degeneration and uncontrolled scale variation.

The TDKV module is designed to strengthen text guidance under weak instance-level semantics. It constructs domain-perturbed textual key-value pairs and learns robust text representations with an asymmetric visibility mask. During cross-modal fusion, the noisy textual keys and values cooperate with clean text as negative samples and contrastive templates for robust semantic guidance.

In summary, the main contributions are as follows.

1) We propose LMLNet for remote sensing SDGOD and curate M^{3}VL with unified annotations, textual descriptions, and cross-domain evaluation settings.
2) The AccA is introduced to represent cross-modal feature structure with SPD matrices and replace fixed inner-product similarity with the AIRM metric. We theoretically prove the affine invariance and stability of AccA, showing that its AIRM-based matching can reduce the bias induced by fixed Euclidean matching.
3) The GPB is introduced to preserve inter-layer geometric structure via structure-preserving bilinear mapping and Stiefel-constrained optimization, thereby preventing feasible-set violation during forward propagation and gradient updates.
4) The TDKV is introduced to generate semantically noisy key-value pairs from domain attribute sets under random perturbations, which provide negative textual samples and contrastive templates for low-information remote sensing vision-language scenarios.

II. RELATED WORK

A. Cross-Domain Object Detection in Remote Sensing

Due to shifts across sensors, scenes, and imaging conditions, detectors trained on one domain often degrade when applied to another. Existing cross-domain detection methods mainly rely on feature alignment, pseudo-label learning, or external knowledge guidance. PiV-AHPC [16] uses local attention to align features and adaptive channel selection to integrate complementary information. DualDA-Net [17] reduces domain discrepancy through coarse-to-fine feature alignment and dual-head co-training. Remote Sensing Teacher [18] introduces a DETR-based adaptation framework with learnable frequency-enhanced feature alignment and self-adaptive pseudo-label assignment. VLSDA [8] uses a frozen VLM as a stable semantic domain, where tridomain prototypical and global contrastive alignment are performed to reduce dependence on noisy target-domain features. However, these cross-domain detection methods usually access unlabeled or sparsely labeled target-domain data during training. In contrast, domain generalization does not use target-domain samples during training and therefore requires stronger robustness to unseen domain shifts.

B. Language-Guided Cross-Domain Detection

Language provides relatively stable semantic priors for cross-domain detection by encoding object categories, attributes, and domain conditions. Category cues preserve object identity, while attribute cues help distinguish stable semantics from domain-dependent appearance. Without target-domain images, aligning these cues with visual regions reduces reliance on source-specific appearance and supports robust detection under domain shift [6], [19], [22].

Existing studies use language for semantic augmentation, style transfer, or feature learning. CLIP the Gap [6] introduces textual domain prompts, and PGST [19] uses GLIP [20] to guide object-level style transfer. LGFR [22] and Boost

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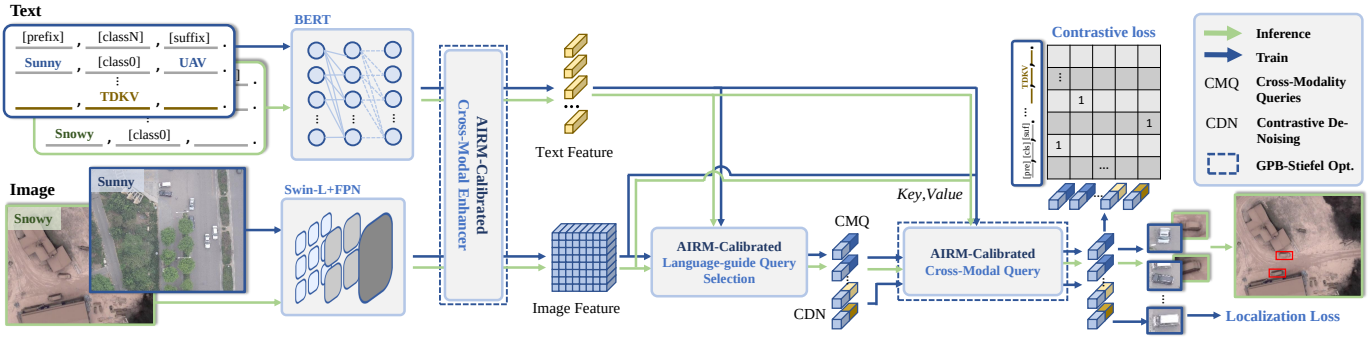


Fig. 2. Schematic of the LMLNet architecture. The training and inference pipelines are largely consistent, except that TDKV-related branches are used only during training.

[7] perform language-guided feature remapping and region-level cross-modal interaction, respectively. In remote sensing, VLSDA [8] aligns visual features with a vision-language semantic space, whereas GOOD [23] uses CLIP-guided style hallucination [21] for domain-generalized oriented detection. However, existing methods often rely on fixed Euclidean metrics and limited or manually designed prompts, restricting their ability to model second-order geometry. Some methods also require target-domain images or overlook geometric feasibility during propagation. These limitations motivate geometry-aware matching, structure-preserving propagation, and noise-robust textual guidance.

C. SPD Manifold-Based Feature Modeling

Structured second-order representations and Riemannian geometric learning model feature correlations in non-Euclidean spaces, offering a geometry-aware alternative for metric and representation learning. SPDGNN [24] constructs GNNs directly on the SPD manifold with the Log-Cholesky metric, using closed-form SPD aggregation and activation to avoid repeated log-exp mappings. SPDDML [25] extends deep metric learning to the SPD manifold for bearing fault diagnosis by integrating a denoising autoencoder, sparsity regularization, and multi-class N-pair loss. SPD-Updater [26] represents templates and search regions with covariance-based SPD descriptors, and measures their similarity using Log-Euclidean geodesic distance for reliable online template updating. RSF [27] performs spatial filtering and dimensionality reduction directly on the SPD manifold to improve robustness in MI-BCI. These studies establish covariance-based SPD representation and Riemannian learning as effective geometric tools. However, they mainly use SPD matrices as task features or learn transformations and distances within a single representation space, without constructing a joint image-text metric for token-level cross-attention under domain shift.

III. METHOD

The following subsections present the overall framework, cross-domain perturbation model, main components, and theoretical analysis of LMLNet.

A. Overview

Fig. 2 illustrates the training and inference pipeline of the proposed method. Text descriptions, together with their TDKV-perturbed variants during training, are encoded by BERT [49], while remote sensing images are processed by Swin-L with FPN to extract multi-scale visual features.

To reduce matching bias caused by fixed Euclidean metrics under domain shift, AccA is incorporated into multiple image-text interaction stages, forming AIRM-Calibrated Cross-Modal Enhancer (ACE), AIRM-Calibrated Language-guided Query Selection (ALQS), and AIRM-Calibrated Cross-Modal Query (ACQ). Specifically, ACE calibrates bidirectional image-text feature enhancement on the SPD manifold. ALQS selects cross-modal queries by modeling the geometric relations between textual semantics and visual regions, while ACQ further updates these queries under an AIRM-calibrated shared metric for robust fusion and localization. To prevent the shared SPD metric state from becoming infeasible during propagation, GPB applies structure-preserving bilinear mapping and Stiefel-constrained optimization within ACE and ACQ. To strengthen text guidance under weak instance-level semantics, TDKV constructs domain-perturbed textual key-value pairs and employs an asymmetric visibility mask to learn robust textual representations during training.

B. Local Affine Perturbation Model

Domain discrepancies in remote sensing imagery are often caused by imaging chains, resolution changes, weather degradation, and scene changes, leading to local geometric variations in feature space [29]. Accordingly, cross-domain variation is modeled as a LAP:

$$x' \approx A_\Omega x + b_\Omega + \varepsilon_\Omega, \tag{1}$$

where $x, x' \in \mathbb{R}^d$ are local features from the source and target domains, A_Ω is a locally valid linear operator. LAP provides a local first-order surrogate for the dominant structured component of cross-domain variation within corresponding feature neighborhoods, while ε_Ω accounts for noise and higher-order effects. A detailed derivation is provided in Appendix A.

However, A_Ω does not in general guarantee invertible information compression or transmission. Therefore, cases

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involving information folding, rank-deficient Jacobians, or ill-conditioned mappings must also be considered [33], which are further analyzed in Section III-E.

C. AIRM-Calibrated Cross-Attention

Standard self-attention and cross-attention [12] compute similarity with Euclidean inner products $q^\top k$, implicitly assuming a fixed metric across domains. To preserve comparability under domain shift, we replace the fixed dot product with a shared-metric inner product, where the shared SPD metric $\Sigma_0 \in \mathcal{S}_{++}^d$ is obtained from the image and text SPD descriptors under AIRM. With Σ_0, the fixed dot product is replaced by an induced inner product, allowing scale and correlation shifts to be absorbed into the metric.

Let $X_{img} \in \mathbb{R}^{n_v \times d}$ and $X_{text} \in \mathbb{R}^{n_t \times d}$ denote the image and text token features, respectively. For a token feature matrix $X \in \mathbb{R}^{n \times d}$, we define the mean vector and centered feature matrix as $\mu(X) = \frac{1}{n}\mathbf{1}^\top X, \quad C(X) = X - \mathbf{1}\mu(X)$, where $\mathbf{1} \in \mathbb{R}^{n \times 1}$. The SPD representation induced by second-order statistics is given by the Centered SPD Construction (CSC):

$$\Sigma(X) = \frac{1}{n} C(X)^T C(X) + \varepsilon I \in \mathcal{S}_{++}^d, \tag{2}$$

where $\varepsilon > 0$ is a spectral lower-bound regularizer. Since $\frac{1}{n}C(X)^\top C(X) \succeq 0$, adding εI ensures $\lambda_{\min}(\Sigma(X)) \geq \varepsilon$, where $\lambda_{\min}(\cdot)$ denotes the smallest eigenvalue. The text and image SPD matrices are then obtained through centered regularization as $\Sigma_{text} = \Sigma(X_{text})$ and $\Sigma_{img} = \Sigma(X_{img})$, respectively. The AIRM [15] metric is employed on $\mathcal{S}_{++}^d$. At the point Σ, the inner product of two tangent vectors U and V in the $T_\Sigma \mathcal{S}_{++}^d$ is defined as follows:

$$g_\Sigma^{AIRM}(U, V) = \frac{1}{2} \operatorname{tr}\left(\Sigma^{-1} \mathrm{U} \Sigma^{-1} \mathrm{V}\right). \tag{3}$$

The corresponding geodesic distance [15] induced by the AIRM inner product is given by:

$$d_{\mathrm{AIRM}}(\Sigma_1, \Sigma_2) = \left\| \log\left(\Sigma_1^{-\frac{1}{2}} \Sigma_2 \Sigma_1^{-\frac{1}{2}}\right) \right\|_F. \tag{4}$$

Where $\|\cdot\|_F$ denotes the Frobenius norm. The cross-modal shared metric can be defined as the weighted Fréchet mean:

$$\begin{aligned} \Sigma_0 = \arg \min_{\Sigma \in \mathcal{S}_{++}^d} \Big[& \alpha\, d_{\mathrm{AIRM}}^2(\Sigma, \Sigma_{\mathrm{img}}) \\ & + (1-\alpha)\, d_{\mathrm{AIRM}}^2(\Sigma, \Sigma_{\mathrm{text}}) \Big], \quad \alpha \in [0,1]. \end{aligned} \tag{5}$$

In the image-text two-point weighted case, the minimizer lies on the geodesic connecting the two and thus admits the following closed-form expression:

$$\begin{aligned} \Sigma_0 &= \Sigma_{img} \#_{1-\alpha} \Sigma_{text} \\ &= \Sigma_{img}^{\frac{1}{2}} \left(\Sigma_{img}^{-\frac{1}{2}} \Sigma_{text} \Sigma_{img}^{-\frac{1}{2}}\right)^{1-\alpha} \Sigma_{img}^{\frac{1}{2}}, \end{aligned} \tag{6}$$

where $\#$ denotes geodesic interpolation under AIRM, and the matrix power $(\cdot)^t$ is defined via spectral calculus. To further inject the shared metric Σ_0 into the matching operator, a whitening transform induced by this metric is introduced and applied to both modal queries:

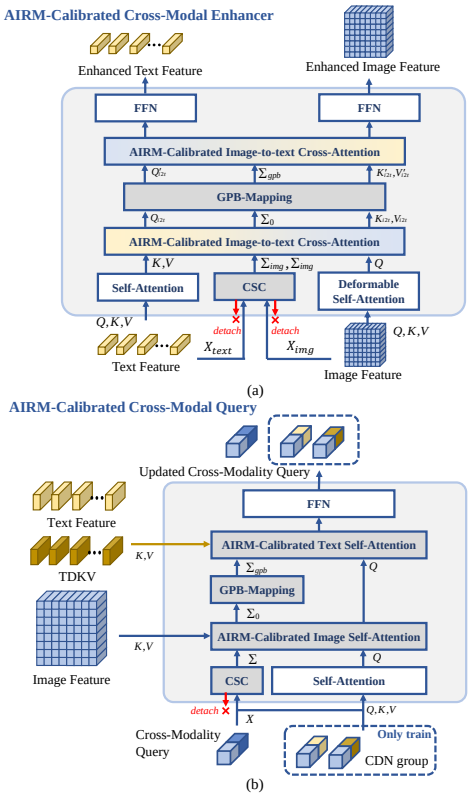


Fig. 3. Schematic illustrations of ACE and ACQ modules. GPB is embedded in these modules and will be introduced in Section III-D.

$$\begin{aligned} W &= \Sigma_0^{-\frac{1}{2}} \\ &= (\Sigma_{img}^{\frac{1}{2}} (\Sigma_{img}^{-\frac{1}{2}} \Sigma_{text} \Sigma_{img}^{-\frac{1}{2}})^{1-\alpha} \Sigma_{img}^{\frac{1}{2}})^{-\frac{1}{2}} \end{aligned} \tag{7}$$

$$\tilde{X}_{img} = X_{img} W, \qquad \tilde{X}_{text} = X_{text} W, \tag{8}$$

where W denotes the whitening transform induced by the shared metric Σ_0, $\tilde{X}_{img}$ and $\tilde{X}_{text}$ represent the whitened image and text query features, respectively.

The original self-attention logits are replaced by the inner product under the shared metric, given by:

$$AccA(Q, K, V) = s\left(\frac{(QW)(KW)^\top}{\sqrt{d}}\right) V \tag{9}$$

$$= s\left(\frac{Q\Sigma_0^{-1} K^\top}{\sqrt{d}}\right) V. \tag{10}$$

Where $s(\cdot)$ denotes the softmax function. AccA replaces the Euclidean dot product $q^\top k$ with the shared-metric inner product $\langle q, k \rangle_{\Sigma_0} \triangleq q^\top \Sigma_0^{-1} k$. As shown in Fig. 3(a), AccA can be used for both text-guided image feature enhancement and image-guided text feature enhancement, forming the ACE.

Grounding DINO [9] employs Language-Guided Query Selection to select cross-modality queries and a Cross-Modality Decoder to refine them. Like the Feature Enhancer, both stages rely on fixed Euclidean inner products and remain susceptible to metric bias under domain shift. We therefore incorporate AccA into their matching operations, forming ALQS and ACQ, which together with ACE maintain AIRM-calibrated

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matching throughout the detection pipeline. In ALQS, image and text features are mapped into the shared metric space by the whitening transform W, and the top N_q image queries are selected for subsequent decoding:

$$I_{N_q}^{\mathrm{AIRM}} = \mathrm{Top}_{N_q}\Big(\mathrm{Max}^{(-1)}\left(X_{\mathrm{img}}\, WW^\top\, X_{\mathrm{text}}^\top\right)\Big) \tag{11}$$

$$= \mathrm{Top}_{N_q}\Big(\mathrm{Max}^{(-1)}\left(X_{\mathrm{img}}\, \Sigma_0^{-1}\, X_{\mathrm{text}}^\top\right)\Big). \tag{12}$$

Where $\mathrm{Max}^{(-1)}$ denotes the maximum operation along the last dimension. As shown in Fig. 3(b), ACQ updates cross-modal queries under the shared SPD metric, thereby reducing attention mismatch caused by domain shift.

Overall, AccA constructs a feature-dependent joint image-text metric from cross-modal second-order statistics through AIRM geodesic interpolation. Its inverse is incorporated into token-level attention in Equation (10). The same calibration is applied to ACE, ALQS, and ACQ for geometry-aware feature enhancement, query selection, and query refinement. Unlike existing covariance-based representation and Riemannian learning methods [13], [24]–[27], AccA neither treats covariance matrices as final task descriptors nor directly uses pairwise SPD distances as attention scores. Instead, SPD geometry serves as an intermediate shared metric that governs cross-modal token matching.

D. GPB and Stiefel-Constrained Optimization

Inspired by [13], we optimize the GPB parameter W_{gpb} on the Stiefel manifold to ensure stable metric propagation.

1) *GPB:* Given the current metric state $\Sigma \in \mathcal{S}_{++}^{d}$, GPB is defined as:

$$\Sigma_{gpb} = \Phi_{W_{gpb}}(\Sigma) := W_{gpb}\,\Sigma\,(W_{gpb})^\top, \tag{13}$$

where $W_{gpb} \in \mathbb{R}^{d'\times d}$ is a learnable matrix, d and $d^{'}$ denote the matrix dimensions of the input and output SPD metric states. In our implementation, we set $d' = d$. If $\Sigma \succ 0$ and W_{gpb} has full row rank, then $\Sigma_{gpb} \succ 0$, thereby ensuring strict preservation of the SPD structure during inter-layer propagation. For $\forall z \neq 0$, the result follows directly from the quadratic form:

$$z^\top \Sigma_{gpb} z = \Big((W_{gpb})^\top z\Big)^\top \Sigma \Big((W_{gpb})^\top z\Big) > 0. \tag{14}$$

Since $\Sigma \succ 0$, the strict inequality holds if $W_{gpb}^\top z \neq 0$ for all $z \neq 0$. This condition is guaranteed by the full row rank of W_{gpb}, or equivalently the full column rank of $W_{gpb}^\top$, which is enforced by the Stiefel constraint. To propagate the geometric state between bidirectional cross-attention in ACE and ACQ, we introduce a shared SPD state chain. For text-to-image interaction followed by image-to-text interaction, the process is written as:

$$AccA_{t2i}(Q_i, K_t, V_t, \Sigma_0) = s\left(\frac{Q_i \Sigma_0^{-1} K_t^\top}{\sqrt{d}}\right) V_t, \tag{15}$$

$$\Sigma_{gpb} = \Phi_{W_{gpb}}(\Sigma_0) = W_{gpb}\Sigma_0 W_{gpb}^\top \in \mathcal{S}_{++}^{d'}, \tag{16}$$

$$AccA_{i2t}(Q_t, K_i, V_i, \Sigma_{gpb}) = s\left(\frac{Q_t \Sigma_{gpb}^{-1} K_i^\top}{\sqrt{d}}\right) V_i. \tag{17}$$

This design keeps the metric state on the SPD manifold during intra-layer propagation, providing consistent and learnable metric support for bidirectional cross-attention.

2) *Stiefel-Constrained Optimization:* The stability of GPB requires W_{gpb} to satisfy well-conditioned geometric constraints during training. Otherwise, scale imbalance in W_{gpb} may be propagated to the SPD metric state through the bilinear mapping, causing numerical instability. Inspired by [31], [32], we constrain W_{gpb} on the compact Stiefel manifold:

$$\mathcal{ST}(d', d) = \{W_{gpb} \in \mathbb{R}^{d'\times d} \mid W_{gpb} W_{gpb}^\top = I_{d'}\}. \tag{18}$$

This constraint guarantees the full row rank of W_{gpb} and suppresses unconstrained scale growth, keeping the mapping parameter within the feasible set. During gradient descent, the Euclidean gradient of GPB with respect to W_{gpb} is obtained by standard backpropagation. Let $G := \mathrm{sym}(\partial L/\partial \Sigma_{gpb})$ be the symmetrized upstream gradient, where $\Sigma_{gpb} = W_{gpb}\Sigma W_{gpb}^\top$ and $\mathrm{sym}(A) = (A + A^\top)/2$. The differential of Σ_{gpb} can be written as:

$$d\Sigma_{gpb} = dW_{gpb}\Sigma W_{gpb}^\top + W_{gpb}\Sigma dW_{gpb}^\top. \tag{19}$$

The first-order variation of the loss can be written as:

$$\begin{aligned} dL &= \langle G,\ d\Sigma_{gpb}\rangle = \mathrm{tr}\big(G^\top d\Sigma_{gpb}\big) \\ &= \mathrm{tr}\big(G^\top dW_{gpb}\Sigma W_{gpb}^\top\big) + \mathrm{tr}\big(G^\top W_{gpb}\Sigma dW_{gpb}^\top\big) \\ &= 2\ \mathrm{tr}\Big((GW_{gpb}\Sigma)^\top dW_{gpb}\Big) \\ &= 2\,\langle GW_{gpb}\Sigma,\ dW_{gpb}\rangle, \end{aligned} \tag{20}$$

where $\langle A, B\rangle = \mathrm{tr}(A^\top B)$ denotes the Frobenius inner product. Therefore, the Euclidean gradient with respect to W_{gpb} is expressed as $\nabla L_{W_{gpb}} = 2\,G\,W_{gpb}\,\Sigma$. To preserve the semi-orthogonal constraint in Equation (18), at the t-th iteration, we project the Euclidean gradient $\nabla L_{W_{gpb}}^{(t)}$ obtained by back-propagation onto the tangent space of the Stiefel manifold at W_{gpb}^t, where $W_{gpb}^t (W_{gpb}^t)^\top = I_{d'}$:

$$\widetilde{\nabla} L_{W_{gpb}}^{(t)} = \nabla L_{W_{gpb}}^{(t)} - \mathrm{sym}\Big(\nabla L_{W_{gpb}}^{(t)} \left(W_{gpb}^t\right)^\top\Big) W_{gpb}^t. \tag{21}$$

This projected direction satisfies $\widetilde{\nabla} L_{W_{gpb}}^{(t)} (W_{gpb}^t)^\top + W_{gpb}^t (\widetilde{\nabla} L_{W_{gpb}}^{(t)})^\top = 0$, and thus preserves $W_{gpb} W_{gpb}^\top = I_{d'}$ in the first-order approximation. The parameter is then updated along this projected direction and retracted back to the Stiefel manifold:

$$W_{gpb}^{t+1} = \Gamma\Big(W_{gpb}^t - \eta\, \widetilde{\nabla} L_{W_{gpb}}^{(t)}\Big), \tag{22}$$

where η is the learning rate, and $\Gamma(\cdot)$ denotes the QR-based retraction operator [31], [32]. This retraction restores the semi-orthogonal feasibility of W_{gpb} after each update.

E. Invariance and Stability Analysis of AccA

To establish the theoretical foundation of ACE and GPB under domain shift, we analyze two LAP settings:

1) In the ideal case, A_Ω is invertible, and we examine whether AIRM preserves strict geometric consistency and cross-domain comparability.

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2) In the general case, A_Ω may become non-invertible due to compression, information folding, rank-deficient Jacobians, or ill-conditioned mappings [33], and we examine whether AIRM geometry remains stable when strict invariance no longer holds.

Proposition 1. *AIRM Inner-Product Invariance on the SPD Manifold Under Invertible A_Ω*

Proof. Let $\mathcal{S}_{++}^d$ denote the manifold of $d \times d$ SPD matrices. For any invertible matrix $A_\Omega \in GL(d)$, the congruence action under the ideal LAP setting is defined as:

$$\phi_{A_\Omega} : \mathcal{S}_{++}^d \to \mathcal{S}_{++}^d, \qquad \phi_{A_\Omega}(\Sigma) = A_\Omega \Sigma A_\Omega^\top. \tag{23}$$

The invertibility of A_Ω ensures that the congruence action preserves positive definiteness. Specifically, for any $z \in \mathbb{R}^d$ with $z \neq \mathbf{0}$, $A_\Omega^\top z \neq \mathbf{0}$, and hence $z^\top A_\Omega \Sigma A_\Omega^\top z = (A_\Omega^\top z)^\top \Sigma (A_\Omega^\top z) > 0$.

For any tangent vectors $U, V \in T_\Sigma \mathcal{S}_{++}^d$, their induced tangent vectors are obtained by differentiating the congruence action along the corresponding tangent directions.

$$\begin{aligned}
&(d\phi_{A_\Omega})_\Sigma[U] \\
&= \frac{d}{dt}\phi_{A_\Omega}(\Sigma + tU)\Big|_{t=0} = \frac{d}{dt} A_\Omega(\Sigma + tU)A_\Omega^\top\Big|_{t=0} \\
&= A_\Omega U A_\Omega^\top,
\end{aligned} \tag{24}$$

Applying the same differentiation procedure to the tangent vector V:

$$(d\phi_{A_\Omega})_\Sigma[V] = A_\Omega V A_\Omega^\top. \tag{25}$$

On $\mathcal{S}_{++}^d$, the AIRM inner product at Σ is defined as:

$$g_\Sigma^{\mathrm{AIRM}}(U, V) \triangleq \frac{1}{2} \operatorname{tr}\left(\Sigma^{-1} U \Sigma^{-1} V\right). \tag{26}$$

Under the ideal LAP setting, Proposition 1 amounts to proving that:

$$g^{\mathrm{AIRM}}_{\phi_{A_\Omega}(\Sigma)}\left((d\phi_{A_\Omega})_\Sigma[U], (d\phi_{A_\Omega})_\Sigma[V]\right) = g_\Sigma^{\mathrm{AIRM}}(U, V) \tag{27}$$

for any $\Sigma \in \mathcal{S}_{++}^d$ and $U, V \in T_\Sigma \mathcal{S}_{++}^d$. Substituting Equations (24) and (25) into the left-hand side gives:

$$\begin{aligned}
&g^{\mathrm{AIRM}}_{A_\Omega \Sigma A_\Omega^\top}\left(A_\Omega U A_\Omega^\top, A_\Omega V A_\Omega^\top\right) \\
&= \frac{1}{2} \operatorname{tr}\left((A_\Omega \Sigma A_\Omega^\top)^{-1}(A_\Omega U A_\Omega^\top)(A_\Omega \Sigma A_\Omega^\top)^{-1}(A_\Omega V A_\Omega^\top)\right) \\
&= \frac{1}{2} \operatorname{tr}\left(A_\Omega^{-\top}\Sigma^{-1}A_\Omega^{-1}(A_\Omega U A_\Omega^\top)A_\Omega^{-\top}\Sigma^{-1}A_\Omega^{-1}(A_\Omega V A_\Omega^\top)\right) \\
&= \frac{1}{2} \operatorname{tr}\left(A_\Omega^{-\top}\Sigma^{-1} U A_\Omega^\top A_\Omega^{-\top}\Sigma^{-1} V A_\Omega^\top\right) \\
&= \frac{1}{2} \operatorname{tr}\left(A_\Omega^{-\top}\Sigma^{-1} U \Sigma^{-1} V A_\Omega^\top\right) \\
&= \frac{1}{2} \operatorname{tr}\left(\Sigma^{-1} U \Sigma^{-1} V\right) \\
&= g_\Sigma^{\mathrm{AIRM}}(U, V).
\end{aligned} \tag{28}$$

The penultimate equality uses the cyclic invariance of the trace:

$$\operatorname{tr}\left(A_\Omega^{-\top} X A_\Omega^\top\right) = \operatorname{tr}\left(X A_\Omega^\top A_\Omega^{-\top}\right) = \operatorname{tr}(X), \tag{29}$$

where $X = \Sigma^{-1} U \Sigma^{-1} V$. Therefore, the AIRM inner product remains strictly invariant under the transformation induced by invertible A_Ω in the ideal LAP setting. □

Theorem 1. *Lipschitz Stability of the Regularized AIRM Inner Product Under Non-Invertible A_Ω*

Under the LAP model, if A_Ω is non-invertible with $\operatorname{rank}(A_\Omega) < d$, then $\phi_{A_\Omega}(\Sigma) = A_\Omega \Sigma A_\Omega^\top$ is generally positive semidefinite and may leave $\mathcal{S}_{++}^d$. Consequently, the inverse, inverse square root, and logarithmic map in AIRM may become ill-defined. To avoid this degeneration, we introduce spectral lower-bound regularization and define:

$$\phi_{A_\Omega,\varepsilon}(\Sigma) \triangleq A_\Omega \Sigma A_\Omega^\top + \varepsilon I, \qquad \varepsilon > 0. \tag{30}$$

For any $z \in \mathbb{R}^d$ with $z \neq \mathbf{0}$, the following holds:

$$\begin{aligned}
&z^\top \phi_{A_\Omega,\varepsilon}(\Sigma) z \\
&= (A_\Omega^\top z)^\top \Sigma (A_\Omega^\top z) + \varepsilon \|z\|_2^2 \geq \varepsilon \|z\|_2^2 > 0.
\end{aligned} \tag{31}$$

Hence, $\phi_{A_\Omega,\varepsilon}(\Sigma) \in \mathcal{S}_{++}^d$, $\lambda_{\min}(\phi_{A_\Omega,\varepsilon}(\Sigma)) \geq \varepsilon$, and $\|\phi_{A_\Omega,\varepsilon}(\Sigma)^{-1}\|_2 \leq \varepsilon^{-1}$, where $\lambda_{\min}(\cdot)$ and $\|\cdot\|_2$ denote the minimum eigenvalue and spectral norm, respectively. Therefore, the AIRM inner product at the regularized point $\phi_{A_\Omega,\varepsilon}(\Sigma)$ is well defined. For any $P \in \mathcal{S}_{++}^d$, the AIRM inner product is defined as:

$$g_P^{\mathrm{AIRM}}(U, V) \triangleq \frac{1}{2} \operatorname{tr}\left(P^{-1} U P^{-1} V\right), \tag{32}$$

where $U, V \in T_P \mathcal{S}_{++}^d$. To analyze the stability bound under non-invertible domain shift, let $A_\Omega \in \mathbb{R}^{d \times d}$ be a non-invertible perturbation matrix and $\varepsilon > 0$. For any $\Sigma_1, \Sigma_2 \succeq 0$, the perturbed SPD states are defined as:

$$\begin{aligned}
P_i &= \phi_{A_\Omega,\varepsilon}(\Sigma_i) \\
&= A_\Omega \Sigma_i A_\Omega^\top + \varepsilon I \in \mathcal{S}_{++}^d, \quad i \in \{1, 2\}.
\end{aligned} \tag{33}$$

According to the linear part of Equation (30), the perturbation directions U and V are induced as: $U_{A_\Omega} = A_\Omega U A_\Omega^\top$ and $V_{A_\Omega} = A_\Omega V A_\Omega^\top$. Along $(U_{A_\Omega}, V_{A_\Omega})$, the AIRM inner products at P_1 and P_2 satisfy the following Lipschitz-type bound:

$$\begin{aligned}
&\left|g_{P_1}^{\mathrm{AIRM}}(U_{A_\Omega}, V_{A_\Omega}) - g_{P_2}^{\mathrm{AIRM}}(U_{A_\Omega}, V_{A_\Omega})\right| \\
&\leq \frac{\|A_\Omega\|_2^6}{\varepsilon^3} \|\Sigma_1 - \Sigma_2\|_F \|U\|_F \|V\|_F.
\end{aligned} \tag{34}$$

This bound shows that regularized AIRM geometry remains stable under non-invertible A_Ω, even though strict invariance no longer holds.

Proof. Let $M_i = P_i^{-1}$ for $i \in \{1, 2\}$ and define $\Delta\Sigma = \Sigma_1 - \Sigma_2$, $\Delta P = P_1 - P_2 = A_\Omega \Delta\Sigma A_\Omega^\top$. The inverse difference identity then gives:

$$\begin{aligned}
\Delta M &\triangleq M_1 - M_2 = -M_1 \Delta P M_2, \\
\|\Delta M\|_2 &\leq \|M_1\|_2 \|\Delta P\|_2 \|M_2\|_2.
\end{aligned} \tag{35}$$

Since $P_i = A_\Omega \Sigma_i A_\Omega^\top + \varepsilon I \succeq \varepsilon I$, we have $\|M_i\|_2 \leq \varepsilon^{-1}$. Moreover, by the submultiplicativity of the spectral norm:

$$\begin{aligned}
\|\Delta P\|_2 &= \|A_\Omega \Delta\Sigma A_\Omega^\top\|_2 \\
&\leq \|A_\Omega\|_2^2 \|\Delta\Sigma\|_2 \leq \|A_\Omega\|_2^2 \|\Delta\Sigma\|_F.
\end{aligned} \tag{36}$$

Substituting the above bounds for $\|M_1\|_2$, $\|M_2\|_2$, and $\|\Delta P\|_2$ into the estimate $\|\Delta M\|_2 \leq \|M_1\|_2 \|\Delta P\|_2 \|M_2\|_2$ yields:

$$\|\Delta M\|_2 \leq \frac{\|A_\Omega\|_2^2}{\varepsilon^2} \|\Sigma_1 - \Sigma_2\|_F. \tag{37}$$

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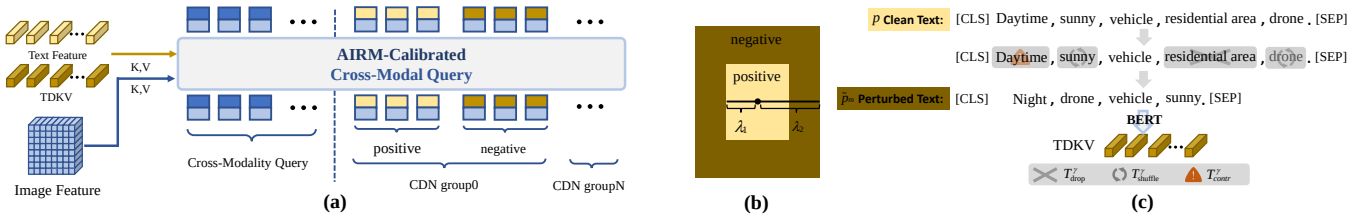


Fig. 4. Illustration of visual-query and textual key-value denoising in ACQ. (a) ACQ with clean and perturbed textual K, V. (b) CDN [34] perturbs ground-truth labels and boxes. (c) TDKV generation pipeline using a simple text example. T^{γ}_{drop} deletes an attribute phrase, $T^{\rho}_{\text{shuffle}}$ reorders attribute phrases, and T^{ξ}_{contr} replaces an attribute with another value from the same group. TDKV is used only during training.

For the induced directions $U_A = A_\Omega U A_\Omega^\top$ and $V_A = A_\Omega V A_\Omega^\top$, we have

$$\|U_A\|_F \le \|A_\Omega\|_2^2 \|U\|_F, \quad \|V_A\|_F \le \|A_\Omega\|_2^2 \|V\|_F. \tag{38}$$

By the AIRM inner product, the value at P_i along U_A and V_A can be written as:

$$g_{P_i}^{\text{AIRM}}(U_A, V_A) = \frac{1}{2} \operatorname{tr}(M_i U_A M_i V_A). \tag{39}$$

Thus, the difference between the AIRM inner products at P_1 and P_2 is given by:

$$\begin{aligned} &g_{P_1}^{\text{AIRM}}(U_A, V_A) - g_{P_2}^{\text{AIRM}}(U_A, V_A) \\ &= \frac{1}{2} \operatorname{tr}\big((M_1 - M_2) U_A M_1 V_A\big) \\ &\quad + \frac{1}{2} \operatorname{tr}\big(M_2 U_A (M_1 - M_2) V_A\big). \end{aligned} \tag{40}$$

Using $|\operatorname{tr}(XY)| \le \|X\|_F \|Y\|_F$ and $\|XY\|_F \le \|X\|_2 \|Y\|_F$, we obtain:

$$\begin{aligned} &\left| g_{P_1}^{\text{AIRM}}(U_A, V_A) - g_{P_2}^{\text{AIRM}}(U_A, V_A) \right| \\ &\le \frac{1}{2} \left(\|M_1\|_2 + \|M_2\|_2 \right) \|\Delta M\|_2 \|U_A\|_F \|V_A\|_F \\ &\le \frac{1}{\varepsilon} \|\Delta M\|_2 \|U_A\|_F \|V_A\|_F. \end{aligned} \tag{41}$$

Substituting Equation (37) and Equation (38) yields

$$\begin{aligned} &\left| g_{P_1}^{\text{AIRM}}(U_A, V_A) - g_{P_2}^{\text{AIRM}}(U_A, V_A) \right| \\ &\le \frac{\|A_\Omega\|_2^6}{\varepsilon^3} \|\Sigma_1 - \Sigma_2\|_F \|U\|_F \|V\|_F. \end{aligned} \tag{42}$$

□

Overall, the above analysis clarifies the role and limitation of regularization in AIRM computation under challenging domain shifts. The bound shows that εI keeps AIRM operations well defined and limits their sensitivity to covariance perturbations. The factor $\|A_\Omega\|_2^6/\varepsilon^3$ further indicates that an appropriate spectral floor ε helps control perturbation amplification and provides numerical protection for AIRM computation. Therefore, the regularization stabilizes AIRM-based metric computation under challenging domain shifts, but cannot reconstruct discriminative target information lost through irreversible feature degradation.

F. Text Denoising Key-Value

In source-only training, prompts combining category and domain attributes may induce spurious associations between object semantics and source-specific conditions, making textual guidance unreliable under unseen domain changes. We therefore select scene, weather, platform, and illumination as representative and observable remote sensing factors and perturb them to construct controlled semantic variations.

Inspired by denoising training [34], we introduce Text Denoising Key-Value (TDKV). As shown in Fig. 4, CDN perturbs ground-truth labels and bounding boxes to construct positive and negative visual queries for improved detection-query learning. In contrast, TDKV applies domain-attribute perturbations to textual (K, V), producing semantic negative counterparts to the clean text and thereby enhancing the discrimination between stable category semantics and domain-dependent variations.

The class-only prompt is retained as a clean anchor during training, while TDKV perturbs only the attribute-aware prompts. Let p denote the clean attribute-aware prompt, and let $\mathcal{C}(\cdot)$ denote a perturbation operator over the predefined domain-attribute set $\mathcal{D}$. During training, the m-th noisy textual key-value prompt is defined as:

$$\tilde{p}_m = \mathcal{C}(p; \mathcal{D}), \quad m = 1, \ldots, N_d, \tag{43}$$

where N_d is the number of generated noisy key-value prompts. As shown in Fig. 4(c), TDKV generates perturbed textual (K, V) from an attribute-aware prompt p. The three perturbation operators are Deletion (T^{γ}_{drop}), Shuffling ($T^{\rho}_{\text{shuffle}}$), and Contradiction ($T^{\xi}_{\text{contr}}$), which delete, reorder, and replace attribute phrases, respectively. These perturbations retain the object category while simulating incomplete, reordered, or conflicting descriptions. Further details are provided in Appendix B.

To prevent noisy key-value tokens from contaminating clean text representations, we apply an asymmetric visibility mask to text-side self-attention. Let L and L_n denote the numbers of clean and noisy tokens, respectively. After placing the clean tokens before the noisy tokens, the attention mask is defined as:

$$M_{\text{text}} = \begin{bmatrix} 0_{L \times L} & -\infty \cdot \mathbf{1}_{L \times L_n} \\ 0_{L_n \times L} & 0_{L_n \times L_n} \end{bmatrix}. \tag{44}$$

This mask allows noisy tokens to attend to clean tokens while preventing clean tokens from accessing noisy ones.

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TABLE I
CROSS-DOMAIN TASK DEFINITIONS IN M^{3}VL

Platform	Symbol	Domain	# Images	# Instances
Satellite	$\mathcal{D}^{\text{sat}}_{\text{air}}$	airport	472	13,577
	$\mathcal{D}^{\text{sat}}_{\text{tra}}$	transportation	1,612	132,467
	$\mathcal{D}^{\text{sat}}_{\text{nat}}$	natural area	1,278	17,755
	$\mathcal{D}^{\text{sat}}_{\text{res}}$	residential area	1,448	75,131
	$\mathcal{D}^{\text{sat}}_{\text{por}}$	port	593	14,991
Drone	$\mathcal{D}^{\text{dro}}_{\text{day}}$	daytime	2,000	30,880
	$\mathcal{D}^{\text{dro}}_{\text{nig}}$	night	2,000	32,463
	$\mathcal{D}^{\text{dro}}_{\text{sno}}$	snow	1,529	2,819
	$\mathcal{D}^{\text{dro}}_{\text{haz}}$	hazy	911	23,545
Camera	$\mathcal{D}^{\text{cam}}_{\text{sun}}$	sunny	2,810	10,508
	$\mathcal{D}^{\text{cam}}_{\text{nig}}$	night	1,098	3,632
	$\mathcal{D}^{\text{cam}}_{\text{rai}}$	rainy	1,299	6,688

Cross-domain task definitions

Task	Shift type	Source	Target	
ill	Illumination	$\mathcal{D}^{\text{dro}}_{\text{day}}$ $\mathcal{D}^{\text{cam}}_{\text{sun}}$	$\mathcal{D}^{\text{dro}}_{\text{nig}}$ $\mathcal{D}^{\text{cam}}_{\text{nig}}$	
wea	Weather	$\mathcal{D}^{\text{dro}}_{\text{day}}$ $\mathcal{D}^{\text{cam}}_{\text{sun}}$	$\mathcal{D}^{\text{dro}}_{\text{sno}}$ $\mathcal{D}^{\text{cam}}_{\text{rai}}$	$\mathcal{D}^{\text{dro}}_{\text{haz}}$
sce	Scene	$\mathcal{D}^{\text{sat}}_{\text{res}}$	$\mathcal{D}^{\text{sat}}_{\text{air}}$ $\mathcal{D}^{\text{sat}}_{\text{nat}}$	$\mathcal{D}^{\text{sat}}_{\text{tra}}$ $\mathcal{D}^{\text{sat}}_{\text{por}}$
pla	Platform	$\mathcal{D}^{\text{dro}}_{\text{day}}$	$\mathcal{D}^{\text{cam}}_{\text{sun}}$ $\mathcal{D}^{\text{sat}}_{\text{tra}}$ $\mathcal{D}^{\text{sat}}_{\text{res}}$	$\mathcal{D}^{\text{sat}}_{\text{air}}$ $\mathcal{D}^{\text{sat}}_{\text{nat}}$ $\mathcal{D}^{\text{sat}}_{\text{por}}$

IV. EXPERIMENTS

A. Dataset

1) *M^{3}VL:* As shown in Fig. 1(a) and Table I, we curate the Multi-Scene, Multi-Weather, Multi-Platform Vision-Language Remote Sensing Dataset (M^{3}VL) for evaluating cross-domain remote sensing object detection. M^{3}VL comprises 12 domain-specific sub-datasets across satellite, drone, and traffic-camera platforms, with 17,050 images and 364,456 object instances in total, while its textual descriptions contain 1,199 unique words.

M^{3}VL is constructed by reorganizing and adapting several public datasets, including xView [41], DIOR [42], and VME [43] for satellite-view object detection; DroneVehicle [44] for drone-based daytime, nighttime, and hazy vehicle detection; NVD [45] for snowy drone imagery; and traffic camera datasets from AAU RainSnow [46] and AI City Challenge [47].

According to practical application scenarios, we define four cross-domain tasks to evaluate model generalization. As shown in Table I, these tasks correspond to illumination, weather, scene, and platform shifts, covering common domain variations in real-world remote sensing applications.

2) *DIOR2HRRSD:* For cross-dataset evaluation, we establish a ten-category setting from DIOR to HRRSD, using the official DIOR [42] as the source domain and the official HRRSD [48] as the target domain. The source domain contains 5,862 images and 28,568 instances, whereas the target domain contains 4,352 images and 8,567 instances. Overall, this setting comprises 10,214 images and 37,135 instances from ten categories: airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle. The associated text corpus contains 1,858,620 tokens and 1,405 unique words.

3) *Construction and quality control:* We first map all vehicle-related labels to a unified category, convert all annotations into a unified bounding-box format, and manually remove duplicate and near-duplicate images. Based on acquisition metadata and visual inspection, we manually divide the remaining images into 12 disjoint domains according to platform, scene, illumination, and weather, as illustrated in Fig. 1(a). Each image belongs to only one domain, and the source and target partitions in Table I contain no overlapping images.

We employ GeoChat-7B [28] to generate hierarchical textual descriptions. Specifically, each GT box is enlarged by factors of 8 and 64, with the resulting regions clipped to the image boundaries. The former describes object color, shape, and size, whereas the latter captures the environment surrounding the object. Each English description component is limited to 30 words. Failed or missing descriptions are manually completed, and all generated text is reviewed by human annotators.

The original differences in platform, sensor, spatial resolution, and scene characteristics are retained. Since these factors may coexist across public data sources, each task name indicates the dominant shift. The reported results primarily reflect generalization under composite domain shifts jointly induced by this dominant factor and its accompanying variations.

B. Implementation Details

1) *Metrics:* For M^{3}VL, we evaluate model generalization on four cross-domain tasks, as shown in Table I, indexed by $m \in \{ill, wea, sce, pla\}$. For each M^{3}VL task m, let $\mathcal{S}^{(m)}$ denote its source-domain set, and let $\mathcal{T}^{(m)}(\mathcal{D}_s)$ denote the target-domain set associated with source domain $\mathcal{D}_s \in \mathcal{S}^{(m)}$.

For each task m, target-domain generalization is summarized by the harmonic mean over all valid source-target pairs:

$$\text{HM}_m = \frac{N_m}{\sum\limits_{\mathcal{D}_s \in \mathcal{S}^{(m)}} \sum\limits_{\mathcal{D}_t \in \mathcal{T}^{(m)}(\mathcal{D}_s)} (P(\mathcal{D}_s \to \mathcal{D}_t) + a)^{-1}}, \quad (45)$$

where $P(\mathcal{D}_s \to \mathcal{D}_t)$ denotes the vehicle-class AP_{50} of a model trained on $\mathcal{D}_s$ and evaluated on $\mathcal{D}_t$, $N_m = \sum_{\mathcal{D}_s \in \mathcal{S}^{(m)}} \left|\mathcal{T}^{(m)}(\mathcal{D}_s)\right|$ is the number of evaluated source-target pairs, and $a = 10^{-6}$ is used for numerical stability.

To reflect the most challenging transfer case in each task, we report the worst-domain AP_{50} as:

$$\text{WO}_m = \min_{\mathcal{D}_s \in \mathcal{S}^{(m)},\ \mathcal{D}_t \in \mathcal{T}^{(m)}(\mathcal{D}_s)} P(\mathcal{D}_s \to \mathcal{D}_t). \quad (46)$$

Performance consistency across the evaluated target domains is quantified by the standard deviation:

$$\text{STD}_m = \Big[\frac{1}{N_m} \sum_{\substack{\mathcal{D}_s \in \mathcal{S}^{(m)} \\ \mathcal{D}_t \in \mathcal{T}^{(m)}(\mathcal{D}_s)}} (P(\mathcal{D}_s \to \mathcal{D}_t) - \text{MEA}_m)^2\Big]^{1/2}, \quad (47)$$

$$\text{MEA}_m = \frac{1}{N_m} \sum_{\substack{\mathcal{D}_s \in \mathcal{S}^{(m)} \\ \mathcal{D}_t \in \mathcal{T}^{(m)}(\mathcal{D}_s)}} P(\mathcal{D}_s \to \mathcal{D}_t). \quad (48)$$

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TABLE II
DISENTANGLED CONTRIBUTIONS OF MANIFOLD MODELING AND ROBUST LANGUAGE GUIDANCE ON THE M^{3}VL DATASET

AccA	GPB	TDKV	$\mathrm{HM}_{sce}\uparrow$	$\mathrm{WO}_{sce}\uparrow$	$\mathrm{MEA}_{sce}\uparrow$	$\mathrm{STD}_{sce}\downarrow$
			56.7	43.5	58.7	10.6
✓			60.3	46.5	62.2	10.4
✓	✓		62.1	48.8	63.8	9.7
✓		✓	61.7	48.3	63.7	10.7
✓	✓	✓	**64.6**	**51.0**	**66.0**	**9.1**

For the DIOR2HRRSD cross-dataset evaluation from DIOR to HRRSD, we adopt COCO-style mAP metrics averaged over all categories. mAP is averaged over IoU thresholds from 0.50 to 0.95, while mAP_{50} and mAP_{75} are computed at fixed IoU thresholds. mAP_S, mAP_M, and mAP_L report the performance for small, medium, and large objects, respectively.

2) *Experimental Settings:* Our model is implemented based on Grounding DINO, using Swin-L with FPN as the visual encoder and BERT [49] as the text encoder. The number of queries and maximum text length are set to 900 and 256, respectively. The balancing factor of the weighted Fréchet mean is set to $\alpha = 0.5$, and the spectral lower-bound regularizer is set to $\varepsilon = 10^{-4}$ in both CSC and the regularized mapping to ensure positive definiteness and numerical stability. In ALQS, the number of retained image queries is set to $N_q = 900$. The training losses follow Grounding DINO: the classification branch uses Focal Loss with $\alpha_f = 0.25$, $\gamma_f = 2.0$, and weight 1.0, while the box regression branch uses L1 Loss with weight 5.0. Localization-related and contrastive losses are retained to enhance cross-modal discriminability. TDKV parameters are provided in Appendix B. Training is conducted on 8 NVIDIA RTX 6000 Ada GPUs with a fixed batch size of 16, while inference is performed on a single NVIDIA RTX 6000 PRO GPU.

All methods are retrained on identical source-domain splits under the same training protocol and image preprocessing, using their official pretrained weights. All VLM methods use the same source-domain textual annotations and predefined class vocabulary while retaining their native prompt formats and tokenizers. Specifically, CLIP-based methods use the CLIP byte-pair encoding tokenizer [6], [7], [10], whereas Grounding DINO [9] and LMLNet use the BERT-base-uncased WordPiece tokenizer. During training, for each class c_k $(k = 1, \ldots, K)$, a description is sampled from its source-domain text pool to construct p_k in the form "[prefix], [c_k], [suffix]." These prompts are concatenated as $P_{\text{train}} = p_1.\ p_2.\ \cdots\ p_K.$ The class-only prompt $P_{\text{test}} = c_1.\ c_2.\ \cdots\ c_K.$ is also retained as a clean anchor during training, while TDKV perturbs only the attribute modifiers in P_{train}. At inference, TDKV is removed and the same fixed P_{test} is used for every unseen target domain. No target-domain descriptions, instance annotations, or domain identifiers are used.

C. Ablation Study

1) *Effectiveness of LMLNet:* Table II further distinguishes the contributions of manifold modeling and robust language guidance through controlled comparisons. With the same textual input, AccA improves HM_{sce} from 56.7 to 60.3, isolating the contribution of AIRM-based manifold matching. Adding GPB further increases it to 62.1, demonstrating the benefit of geometry-preserving metric propagation. On the same AccA baseline, TDKV raises the result to 61.7, confirming the contribution of robust textual supervision. Their combination achieves the best result of 64.6, demonstrating their complementarity.

TABLE III
EFFECT OF DIFFERENT ACCA ON THE M^{3}VL DATASET

SPD	Metric	$\mathrm{HM}_{sce}\uparrow$	$\mathrm{WO}_{sce}\uparrow$	$\mathrm{MEA}_{sce}\uparrow$	$\mathrm{STD}_{sce}\downarrow$
×	×	56.7	43.5	58.7	10.6
✓	×	57.9	45.7	59.2	**9.0**
✓	Log-Euclidean	57.4	45.9	59.0	9.2
✓	AIRM	**60.3**	**46.5**	**62.2**	10.4

TABLE IV
EFFECT OF APPLYING ACCA TO DIFFERENT MODULES ON THE M^{3}VL DATASET

ACE	ALQS	ACQ	$\mathrm{HM}_{sce}\uparrow$	$\mathrm{WO}_{sce}\uparrow$	$\mathrm{MEA}_{sce}\uparrow$	$\mathrm{STD}_{sce}\downarrow$
			56.7	43.5	58.7	10.6
✓			58.0	44.1	60.1	10.7
✓	✓		59.1	45.4	61.1	10.6
✓	✓	✓	**60.3**	**46.5**	**62.2**	**10.4**

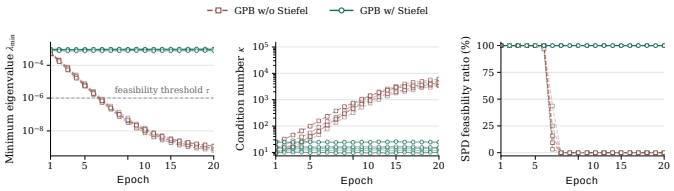


Fig. 5. Effect of the Stiefel constraint on GPB stability: (a) minimum eigenvalue, (b) condition number, and (c) SPD feasibility ratio ($\tau = 10^{-6}$). Each setting is evaluated over five independent runs. GPB w/o Stiefel and GPB w/ Stiefel correspond to the last two rows of Table V.

TABLE V
EFFECT OF SPD STATE PROPAGATION ON THE M^{3}VL DATASET

Setting	$\mathrm{HM}_{sce}\uparrow$	$\mathrm{WO}_{sce}\uparrow$	$\mathrm{MEA}_{sce}\uparrow$	$\mathrm{STD}_{sce}\downarrow$
w/o prop.	60.3	46.5	62.2	10.4
Linear prop.	60.4	45.9	62.4	10.6
GPB w/o Stiefel	61.5	47.1	63.5	10.5
GPB w/ Stiefel	**62.1**	**48.8**	**63.8**	**9.7**

2) *Effectiveness of AccA:* Table III disentangles the contributions of second-order representation and manifold metric geometry. The covariance-only setting increases HM_{sce} from 56.7 to 57.9, whereas AIRM achieves 60.3, confirming that the improvement cannot be explained by covariance encoding alone. To evaluate the cumulative effect of AccA, we progressively incorporate ACE, ALQS, and ACQ into Grounding DINO [9]. As shown in Table IV, HM_{sce} progressively increases from the baseline value of 56.7 to 58.0, 59.1, and 60.3. These consistent gains indicate that fixed Euclidean matching at successive interaction stages limits cross-domain generalization, while applying AccA at each stage provides complementary improvements.

3) *Effectiveness of GPB:* To study structure-preserving propagation and Stiefel-constrained optimization, we conduct ablation experiments on SPD state propagation. As shown in Table V, the best performance is achieved when both designs

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TABLE VI
EFFECT OF TDKV AND VISIBILITY MASK ON THE M^{3}VL DATASET

TDKV	Mask	$\mathrm{HM}_{sce}\uparrow$	$\mathrm{WO}_{sce}\uparrow$	$\mathrm{MEA}_{sce}\uparrow$	$\mathrm{STD}_{sce}\downarrow$
×	×	60.3	46.5	62.2	10.4
✓	×	61.3	46.7	61.5	**10.0**
✓	Symmetric	61.0	47.6	62.9	10.6
✓	Asymmetric	**61.7**	**48.3**	**63.7**	10.7

TABLE VII
EFFECT OF DIFFERENT TDKV PERTURBATIONS UNDER DOMAIN SHIFTS ON THE M^{3}VL DATASET

Perturbation	$\mathrm{HM}_{ill}\uparrow$	$\mathrm{HM}_{wea}\uparrow$	$\mathrm{HM}_{sce}\uparrow$	$\mathrm{HM}_{pla}\uparrow$
Contradiction	71.1	82.0	61.2	52.2
Shuffling	70.7	81.4	60.9	51.1
Deletion	68.9	80.3	59.6	50.3
Combined	**71.3**	**82.3**	**61.7**	**52.6**

TABLE VIII
SENSITIVITY ANALYSIS OF KEY HYPERPARAMETERS ON THE M^{3}VL DATASET. ONE PARAMETER IS VARIED AT A TIME

Parameter	Value	$\mathrm{HM}_{sce}\uparrow$	$\mathrm{WO}_{sce}\uparrow$	$\mathrm{MEA}_{sce}\uparrow$	$\mathrm{STD}_{sce}\downarrow$
α	0.3	61.5	48.1	63.5	10.8
	0.5	**61.7**	**48.3**	**63.7**	**10.7**
	0.7	60.8	47.4	62.8	11.0
ε	10^{-5}	60.5	46.9	62.4	11.2
	10^{-4}	**61.7**	**48.3**	**63.7**	**10.7**
	10^{-3}	61.6	48.0	63.4	10.9
N_d	1	59.4	46.5	60.8	**10.1**
	2	60.2	47.1	62.1	10.7
	4	**61.7**	**48.3**	**63.7**	10.7
	8	60.0	47.5	62.0	10.9

are introduced, indicating their importance for stable SPD state propagation. To investigate the geometric stability of GPB during feature transfer, we record the spectral and numerical properties of its propagated covariance matrix Σ_{gpb}. The figure overlays the five unsmoothed random-seed trajectories with their median and interquartile range. As shown in Fig. 5, the Stiefel-constrained GPB maintains a minimum eigenvalue of 9.385×10^{-4}, a condition number of 1.459×10^{1}, and a numerical SPD validity rate of 100%. Without the Stiefel constraint, these metrics deteriorate to 8.662×10^{-10}, 4.455×10^{3}, and 0%, respectively. These results confirm that the Stiefel-constrained GPB effectively preserves SPD feasibility during metric propagation.

4) *Effectiveness of TDKV:* To evaluate TDKV and its visibility mask, we compare four settings: without TDKV, TDKV without mask, TDKV with a symmetric mask, and TDKV with the proposed asymmetric mask. As shown in Table VI, TDKV improves performance, and the asymmetric mask achieves the best result. We compare three TDKV perturbations and their combination across four domain shifts. Deletion simulates missing attributes to encourage robust representation learning, a motivation shared with self-supervised masked reconstruction [35]. As shown in Table VII, contradiction performs best individually, while combining all three perturbations achieves the best performance across all four shifts, indicating their complementary benefits.

5) *Hyperparameter Sensitivity:* We evaluate the sensitivity of LMLNet by varying one hyperparameter at a time. As shown in Table VIII, performance remains relatively stable across the tested ranges. The results at $\alpha = 0.3$ and $\varepsilon = 10^{-3}$ are close to the defaults, whereas $\alpha = 0.7$ and $\varepsilon = 10^{-5}$ cause moderate degradation. TDKV also remains stable for $N_d = 1$ to 8 and achieves the best overall accuracy at $N_d = 4$. These results demonstrate the robustness of LMLNet to the three hyperparameters within appropriate ranges.

D. Comparisons With Other Detectors

To comprehensively evaluate the proposed method, we compare it with three groups of representative methods. General-purpose object detectors, including Faster R-CNN [36], CenterNet [37], DINO [34], DDQ [38], and YOLO26x [39], are adopted as generic baselines. Single-domain generalized object detectors, including DivAlign [2], OA-DG [3], DG-YOLO [40], CDSD [1], and G-NAS [51], are introduced to evaluate cross-domain competitiveness. Vision-language-assisted methods, including Grounding DINO [9], YOLO-World [10], CLIP the Gap [6], and Boost [7] are included for comparison.

1) *Quantitative Analysis on M^{3}VL:* As shown in Table IX, illumination variation introduces clear domain shifts, under which conventional detectors show limited generalization. Single-domain generalized detectors generally achieve better results than general-purpose object detectors, showing their advantage in cross-illumination generalization. Notably, our method obtains higher HM_{ill}, WO_{ill}, and MEA_{ill}, while yielding lower STD_{ill}. One reason is that illumination variation can be well approximated by the local affine perturbation model in Equation (1).

Weather variation introduces substantial appearance changes across target domains. Notably, the proposed method achieves the best HM_{wea} and MEA_{wea}, demonstrating its superior overall generalization under weather variations. Weather-induced cross-domain perturbations can cause similarity drift under Euclidean inner-product matching. In contrast, AccA models image-text interactions using covariance-based SPD statistics and constructs a shared AIRM metric via the Fréchet mean.

As shown in Table X, cross-scene transfer causes clear performance degradation for most compared methods. The proposed method achieves the best overall performance, with the highest HM_{sce}, WO_{sce}, and MEA_{sce}, and lower STD_{sce}. Scene shifts directly disturb the structural correspondence between visual regions and textual semantics. AccA enables more robust feature enhancement, query selection, and cross-modal query updating under scene variations, leading to superior cross-scene generalization. Under platform shifts, LMLNet achieves the highest HM, WO, and MEA scores, respectively.

2) *Quantitative Analysis on DIOR2HRRSD:* Table XI shows that generic detectors exhibit limited cross-dataset generalization, particularly for small objects. Single-domain generalized and vision-language-assisted methods improve the overall performance but remain less effective under strict localization and scale variations. In contrast, LMLNet achieves the best results across all metrics, demonstrating stronger multi-category generalization and more robust detection across object scales.

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TABLE IX
QUANTITATIVE COMPARISONS UNDER ILLUMINATION AND WEATHER SHIFTS ON THE M^{3}VL DATASET. ↑ INDICATES THAT HIGHER IS BETTER, AND ↓ INDICATES THAT LOWER IS BETTER. THE BEST RESULTS ARE SHOWN IN BOLD RED, AND THE SECOND-BEST RESULTS ARE SHOWN IN BOLD BLACK

Method	Visual Encoder	Illumination				Weather				FPS ↑
		HM_{ill} ↑	WO_{ill} ↑	MEA_{ill} ↑	STD_{ill} ↓	HM_{wea} ↑	WO_{wea} ↑	MEA_{wea} ↑	STD_{wea} ↓	
General-purpose Object Detectors										
Faster R-CNN [36]	Resnext101	40.8	30.0	46.8	16.8	64.0	52.1	67.7	16.7	57
CenterNet [37]	DLA-102	38.0	28.1	43.4	15.3	60.1	49.7	64.1	17.3	72
DINO [34]	Swin-L	36.0	27.3	40.2	12.8	43.8	32.0	47.5	13.4	16
DDQ [38]	Swin-L	34.4	26.7	37.5	10.8	46.4	34.5	49.6	12.5	15
YOLO26x [39]	-	42.1	32.7	45.8	13.1	48.5	39.8	52.8	16.8	**73**
Single-domain Generalized Object Detectors										
DivAlign [2]	Resnet101	57.7	49.2	59.4	10.3	65.8	54.2	67.6	10.6	53
OA-DG [3]	Resnet101	46.1	37.2	48.9	11.7	60.7	50.2	62.4	10.4	44
DG-YOLO [40]	Darknet53	27.3	21.1	29.9	**8.8**	22.4	13.6	26.7	9.3	**84**
CDSD [1]	Resnet101	38.7	32.9	40.0	**7.2**	40.5	27.3	44.8	12.7	41
G-NAS [51]	Swin-L	67.9	58.9	69.6	10.6	71.3	63.4	72.6	9.9	14
Vision-Language Assisted Methods										
Grounding DINO [9]	Swin-L	62.5	52.4	64.9	12.5	75.4	63.5	76.8	10.0	12
YOLO-World-L [10]	-	56.2	46.9	58.6	11.7	68.7	59.4	69.8	**8.7**	18
CLIP the Gap [6]	Swin-L	54.3	44.8	56.9	12.1	61.3	50.2	62.7	9.1	11
Boost [7]	Swin-L	**69.0**	**59.6**	**70.7**	11.1	**80.1**	**71.5**	**80.8**	**7.7**	12
Ours	Swin-L	**72.2**	**63.5**	**73.6**	10.1	**83.0**	**70.8**	**84.2**	9.6	10

TABLE X
QUANTITATIVE COMPARISONS UNDER SCENE AND PLATFORM SHIFTS ON THE M^{3}VL DATASET. ↑ INDICATES THAT HIGHER IS BETTER, AND ↓ INDICATES THAT LOWER IS BETTER. THE BEST RESULTS ARE SHOWN IN BOLD RED, AND THE SECOND-BEST RESULTS ARE SHOWN IN BOLD BLACK

Method	Visual Encoder	Scene				Platform				FPS ↑
		HM_{sce} ↑	WO_{sce} ↑	MEA_{sce} ↑	STD_{sce} ↓	HM_{pla} ↑	WO_{pla} ↑	MEA_{pla} ↑	STD_{pla} ↓	
General-purpose Object Detectors										
Faster R-CNN [36]	Resnext101	13.8	9.3	15.1	4.3	0.5	0.2	1.8	**2.2**	57
CenterNet [37]	DLA-102	10.7	6.1	12.6	**4.2**	0.8	0.4	1.3	**1.2**	72
DINO [34]	Swin-L	42.7	27.4	46.4	11.2	0.7	0.3	2.4	3.3	16
DDQ [38]	Swin-L	40.3	23.6	45.6	12.8	1.0	0.5	2.8	3.5	15
YOLO26x [39]	-	9.6	6.8	10.5	**3.0**	0.9	0.4	2.7	3.3	**73**
Single-domain Generalized Object Detectors										
DivAlign [2]	Resnet101	38.9	29.1	41.8	11.1	33.3	17.2	37.7	10.5	53
OA-DG [3]	Resnet101	35.8	23.5	39.8	12.2	31.8	15.8	36.7	10.9	44
DG-YOLO [40]	Darknet53	23.3	15.9	25.6	7.6	17.7	9.5	20.9	7.7	**84**
CDSD [1]	Resnet101	35.1	24.4	37.6	9.4	13.5	9.2	16.7	9.8	41
G-NAS [51]	Swin-L	59.1	**46.1**	60.8	9.9	42.7	19.2	50.4	14.1	14
Vision-Language Assisted Methods										
Grounding DINO [9]	Swin-L	56.7	43.5	58.7	10.6	30.8	18.0	36.2	12.6	12
YOLO-World-L [10]	-	53.9	39.8	56.0	9.8	29.6	16.0	38.8	14.6	18
CLIP the Gap [6]	Swin-L	42.3	34.6	44.5	10.7	31.6	14.6	37.5	11.4	11
Boost [7]	Swin-L	**59.9**	45.3	**61.9**	10.4	**52.1**	**34.2**	**54.3**	9.4	12
Ours	Swin-L	**64.6**	**51.0**	**66.0**	9.1	**54.7**	**37.3**	**56.5**	8.8	10

3) *Visualization Analysis of Results:* We further examine the feature-level discrepancy under $Task_{pla}$ on M^{3}VL by randomly sampling 500 positive features from each of the drone, camera, and satellite domains before the detection head and projecting them using t-SNE [50]. Since all samples belong to the vehicle category, their domain-wise separation mainly reflects platform-specific bias. As shown in Fig. 6, the compared methods retain distinguishable platform-dependent clusters, whereas LMLNet produces the strongest cross-platform overlap with only a few isolated samples. This result indicates that LMLNet reduces platform-related feature discrepancy and learns more transferable representations.

We further compute the unbiased empirical squared maximum mean discrepancy (MMD^2) [52] using the original high-dimensional positive-sample features. Before computing $\text{MMD}^2(\times 10^2)$, the features are ℓ_2-normalized, and an RBF kernel with $\sigma = 1.0$ is employed. A lower $\text{MMD}^2(\times 10^2)$ value indicates a smaller cross-platform feature-distribution discrepancy. As annotated in Fig. 6, LMLNet achieves the lowest $\text{MMD}^2(\times 10^2)$ values for both domain pairs, quantitatively demonstrating its superior ability to suppress platform-specific bias and learn more transferable representations.

Fig. 7(a) shows representative challenging cases from $Task_{pla}$, $Task_{wea}$, and $Task_{ill}$, covering cross-platform, cross-weather, and cross-illumination shifts. In port scenes, the coexistence of ships and vehicles further causes semantic con-

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TABLE XI
MULTI-CATEGORY CROSS-DATASET COMPARISONS FROM DIOR TO HRRSD. THE BEST AND SECOND-BEST RESULTS ARE SHOWN IN BOLD RED AND BOLD BLACK

Method	mAP	mAP_{50}	mAP_{75}	mAP_S	mAP_M	mAP_L
Generic Detectors						
YOLO26x [39]	28.4	46.5	31.0	2.5	24.9	33.5
Faster R-CNN-X101 [36]	21.8	39.8	21.9	3.4	14.2	29.0
DINO-L [34]	24.0	42.3	24.7	12.5	19.5	33.6
Single-Domain Generalized Detectors						
DivAlign [2]	30.7	49.4	33.2	4.8	26.5	37.1
DG-YOLO [40]	29.5	48.2	31.8	1.2	25.7	35.4
CDSD [1]	31.6	50.9	34.5	5.9	27.8	38.0
Vision-Language Assisted Methods						
Grounding DINO [9]	33.4	52.8	36.7	15.7	28.8	42.2
CLIP the Gap [6]	**34.5**	52.9	**38.4**	16.9	27.9	**45.0**
Boost [7]	33.8	**55.6**	37.1	**17.4**	**31.2**	43.3
LMLNet (Ours)	**38.7**	**59.4**	**42.6**	**20.1**	**34.0**	**47.5**

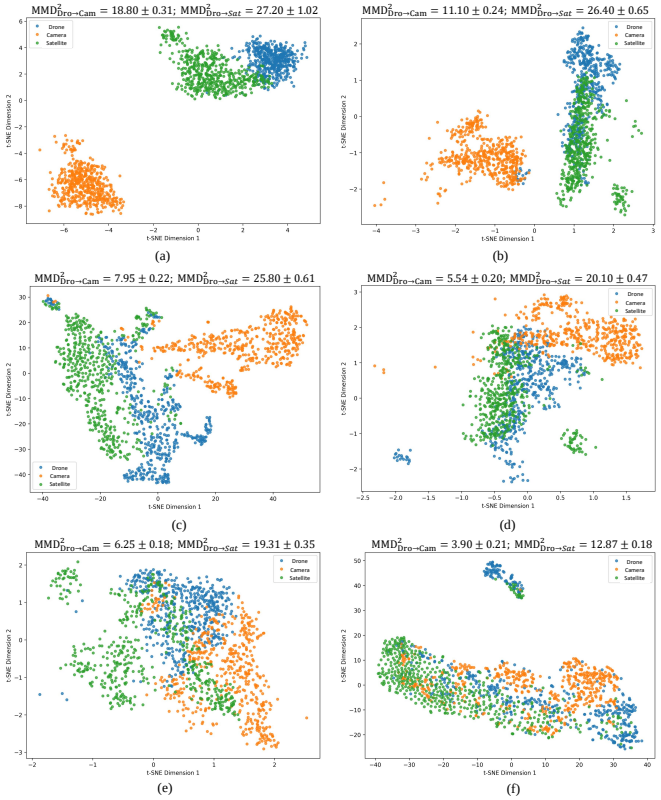


Fig. 6. t-SNE visualization of positive features under cross-platform shifts. Colors denote different platform domains. The values above each subfigure report the mean ± standard deviation of $\mathrm{MMD}^2(\times 10^2)$ over five random feature samplings. Subfigures (a)–(f) correspond to Faster R-CNN [36], DG-YOLO [40], Grounding DINO [9], CDSD [1], CLIP the Gap [6], and LMLNet, respectively.

fusion and incomplete vehicle detection. In contrast, LMLNet produces more complete and accurate results. By calibrating image-text interaction with AccA under the AIRM metric, LMLNet strengthens the correspondence between textual semantics and visual regions, thereby suppressing background interference. Fig. 7(b) further compares different methods on the HRRSD target domain. LMLNet produces more complete detections of multiple airplanes and dense vehicles than the compared methods, qualitatively supporting its multi-category cross-dataset generalization.

TABLE XII
COMPATIBILITY ACROSS FRAMEWORKS ON M^3VL

Configuration	LML	$HM_{ill}\uparrow$	$HM_{wea}\uparrow$	$HM_{sce}\uparrow$	$HM_{pla}\uparrow$	Δ
GDINO (BERT-Base)		62.5	75.4	56.7	30.8	–
	✓	71.0	82.2	**62.1**	**53.3**	**+10.8**
GDINO (CLIP-B/32)		63.0	76.0	55.9	30.0	–
	✓	**73.5**	**85.7**	61.3	35.9	+7.9
YOLO-World (CLIP-B/32)		56.2	68.7	53.9	29.6	–
	✓	66.7	80.8	58.7	35.9	+8.4

TABLE XIII
EMPIRICAL LAP ANALYSIS UNDER CONTROLLED DOMAIN SHIFTS

Condition	Severity	$R_{\mathrm{LAP}}\downarrow$	LMLNet	GDINO	ΔAP_{50}
Haze	Moderate	0.3474 ± 0.0085	91.4	78.9	+12.5
	Severe	0.4451 ± 0.0110	88.2	69.4	+18.8
Snow	Moderate	0.3622 ± 0.0124	92.3	82.7	+9.6
	Severe	0.5580 ± 0.0135	72.7	65.6	+7.1
Resolution	Moderate	0.1671 ± 0.0032	90.5	77.6	+12.9
	Severe	0.5249 ± 0.0027	56.9	51.7	+5.2

E. Compatibility Analysis

We evaluate the compatibility of the proposed manifold strategy across different text encoders and detection frameworks. For a controlled comparison, LML includes AccA and GPB, while TDKV is excluded because YOLO-World does not employ query-based detection. The evaluated configurations cover contextual language pretraining with BERT-Base and image-text contrastive pretraining with CLIP-B/32. As shown in Table XII, Grounding DINO (GDINO) with CLIP-B/32 performs best under illumination and weather shifts, whereas the BERT-Base configuration performs best under scene and platform shifts. Grounding DINO [9] further benefits from box-level detection and phrase-grounding pretraining, providing region-level localization priors for small remote-sensing targets. Nevertheless, LML consistently improves all configurations, indicating that its effectiveness is not primarily determined by a specific detection framework, text encoder, or source of pretrained knowledge.

F. Empirical Applicability Analysis of LAP

Both LMLNet and GDINO are trained only on the daytime drone source domain $\mathcal{D}_{\mathrm{day}}^{\mathrm{dro}}$. Six controlled target domains are generated from its test images, covering moderate and severe haze, snow appearance, and resolution degradation. The atmospheric scattering coefficients for moderate and severe haze are set to $\beta = 2.0$ and $\beta = 3.0$, respectively. The snow levels are synthesized through semantic image editing at two accumulation intensities and manually checked for spatial consistency. The resolution levels are generated through Gaussian prefiltering, area downsampling with scale factors of 0.25 and 0.12, and bicubic reconstruction. Fig. 8 visualizes the source image and the moderate variant of each domain shift.

To assess the approximation accuracy of LAP for cross-domain feature transformations, we compute R_{LAP} from paired RoI features (x_i, x_i') extracted by the visual encoder of LMLNet. Using identical GT boxes, we apply RoIAlign to the multi-scale features of each source image and its target-domain counterpart, followed by spatial pooling and

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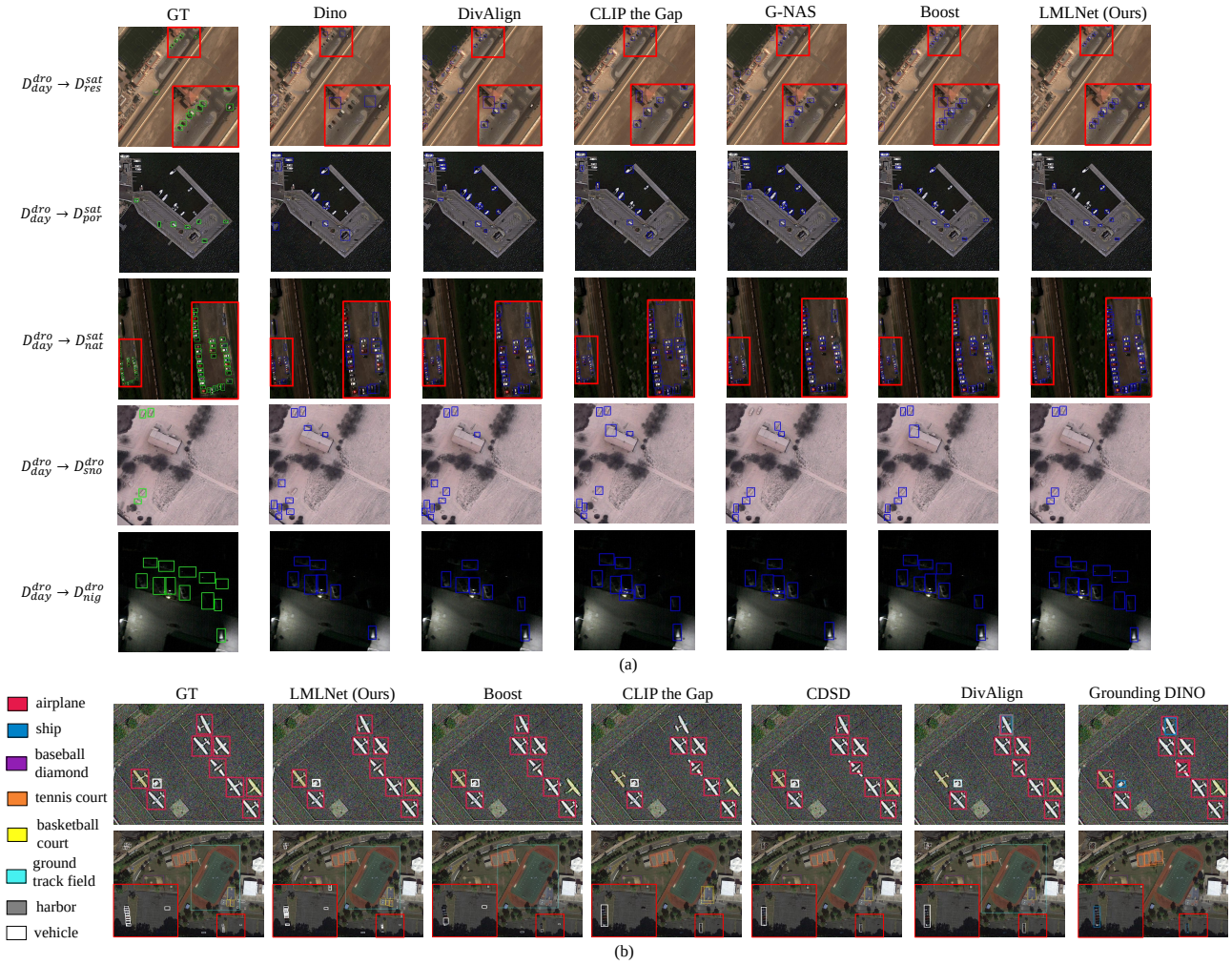


Fig. 7. Qualitative comparisons on M^{3}VL and HRRSD. (a) M^{3}VL: Green and blue boxes denote GT and predictions, respectively, while red rectangles highlight challenging regions. (b) HRRSD: box colors denote object categories, and red rectangles in the second row highlight dense vehicle regions.

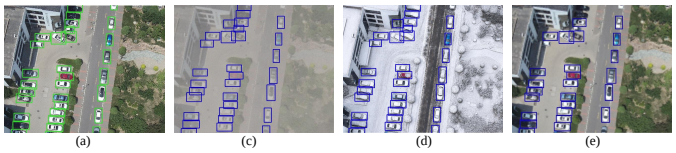


Fig. 8. Visualization under moderate controlled domain shifts. (a) Source-domain image with ground-truth annotations. (b)–(d) LMLNet predictions under Moderate haze, moderate snow appearance, and moderate resolution degradation, respectively. Green and blue boxes denote ground-truth annotations and predictions, respectively.

concatenation. The paired images are divided by image ID into 70% mapping-estimation pairs and 30% held-out pairs. On the former, PCA reduces the feature dimension to 64, the source features are divided into four local clusters, and ridge regression estimates a local affine mapping for each cluster. The normalized prediction residual on the held-out RoIs is calculated as:

$$R_{\mathrm{LAP}} = \frac{1}{N}\sum_{i=1}^{N} \frac{\left\| x_i' - \left(\widehat{A}_{\Omega(i)} x_i + \widehat{b}_{\Omega(i)}\right)\right\|_2}{\|x_i'\|_2 + \delta}, \qquad (49)$$

where $\Omega(i)$ denotes the nearest cluster of the ith source RoI, $\delta = 10^{-6}$, and N is the number of held-out RoIs. We repeat the split five times and report the mean and standard deviation. As shown in Table XIII, LAP remains applicable to weather, seasonal appearance, and resolution variations when local feature correspondence is preserved. In the tested settings, LMLNet consistently outperforms GDINO when $R_{\mathrm{LAP}} < 0.5$, while both methods show marked performance degradation when $R_{\mathrm{LAP}} > 0.5$.

LAP is therefore most applicable to illumination and weather variations, as well as platform shifts with moderate resolution degradation, when local feature correspondence is preserved. Under severe information loss or semantic shifts, $\|\varepsilon_\Omega\|$ increases and the LAP approximation becomes unreliable, as further discussed in Section V. Accordingly, LMLNet primarily targets domain shifts in which object structures and their local spatial relations with the surrounding context remain at least partially preserved.

G. Efficiency Analysis

As shown in Table XIV, we compare LMLNet with DINO, Grounding DINO, G-NAS, and Boost, which serve as representative and informative baselines on M^{3}VL. All models are profiled with 1024×1024 inputs and a batch size of 1

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TABLE XIV
COMPUTATIONAL COMPLEXITY ON M^{3}VL.

Method	Visual Backbone	Params (M)	FLOPs (T)
DINO [34]	Swin-L	261.30	2.104
Grounding DINO [9]	Swin-L	343.12	2.317
G-NAS [51]	Swin-L	292.85	2.177
Boost [7]	Swin-L	325.16	2.231
LMLNet w/o AccA	Swin-L	343.31	2.322
LMLNet w/o GPB	Swin-L	343.72	2.372
LMLNet	Swin-L	343.92	2.379



Fig. 9. Failure cases under target-domain visual degradation. (a) Source-domain image from $\mathcal{D}_{\text{day}}^{\text{dro}}$. (b) Target-domain image from $\mathcal{D}_{\text{res}}^{\text{sat}}$. (c) SDM-Car [53] image. Green boxes denote ground truth and blue boxes denote predictions.

on a single NVIDIA RTX 6000 PRO GPU. Compared with Grounding DINO, LMLNet increases Params by only 0.23% and FLOPs by 2.6%, while improving HM_{sce} and HM_{pla} by 7.9 and 23.9 points, respectively.

Since category and domain prompts are typically fixed or repeatedly used, their BERT embeddings can be pre-computed and cached, removing the text encoder from the online inference path. Under the same setting, training takes approximately 3 and 3.5 hours for Grounding DINO and LMLNet, respectively. The peak inference memory increases from 4194.70 MiB to 4250 MiB, but remains within the capacity of remote sensing edge devices equipped with at least 8 GB of accelerator memory [54].

V. DISCUSSION

1) *Limitations:* Although LMLNet achieves the best overall performance across the four cross-domain tasks, a limitation remains. LMLNet may fail under severe target-feature degradation. As shown in Fig. 9, LMLNet performs well on our target-domain image in Fig. 9(b), where vehicles remain recognizable. In contrast, vehicles in the SDM-Car [53] image in Fig. 9(c) degenerate into small bright points, causing missed detection. This indicates that reliable localization remains challenging when object appearance is extremely weak.

2) *Future Work:* Future work will explore context-aware language guidance and multimodal sensing to address severe target-feature degradation. Roads, parking areas, and inter-building spaces may provide contextual cues when object appearance is weak. Inspired by occlusion-guided fusion [55], we will also investigate combining optical imagery with LiDAR to exploit complementary geometry while suppressing unreliable visual features.

VI. CONCLUSION

This paper introduces a Language-Guided Manifold Learning Network, named LMLNet, for single-domain generalized object detection in remote sensing scenarios. To address the metric bias caused by fixed Euclidean inner-product matching, LMLNet introduces AccA, which constructs a shared SPD metric from image-text second-order statistics and performs cross-modal matching under an AIRM-induced metric. To prevent geometric constraints from being broken during deep propagation, LMLNet employs GPB with Stiefel-constrained optimization, ensuring stable SPD metric propagation. To alleviate weak semantic guidance caused by short and sparse instance-level text, LMLNet introduces TDKV, which injects domain-aware perturbations into textual keys and values to enhance semantic robustness. Experiments on M^{3}VL and DIOR2HRRSD datasets demonstrate the cross-domain generalization of LMLNet, while ablation studies validate its components.

APPENDIX A
DERIVATION OF THE LAP MODEL

1) *Cross-illumination perturbation.* In remote sensing imaging, variations in exposure gain and solar elevation angle can be approximated as local affine radiometric transformations:

$$x' \approx A_\Omega x + b_\Omega, \tag{50}$$

where A_Ω models channel scaling and mixing caused by illumination, contrast, and sensor response variations, and b_Ω represents the bias induced by background radiance changes.

2) *Cross-resolution perturbation.* Inspired by [30], cross-resolution degradation can be modeled as

$$I_{\text{low}} \approx D\left(H * I_{\text{high}}\right) + \eta, \tag{51}$$

where H denotes the convolution kernel associated with the point spread function, $D(\cdot)$ is the resampling or downsampling operator, and η represents additive noise. After vectorizing the image, the degradation model can be written as:

$$i_{low} \approx D_m H_c\, i_{high} + \eta = A\, i_{high} + \eta, \tag{52}$$

where H_c and D_m are the matrix forms of convolution and downsampling, respectively, and $A = D_m H_c$ is the overall degradation operator. Let the feature extractor be a differentiable mapping $f : \mathbb{R}^N \to \mathbb{R}^d$, with $x = f\left(i_{high}\right)$ and $x' = f\left(i_{low}\right)$. Denote the input perturbation by $\Delta = i_{low} - i_{high} \approx (A - I)\, i_{high} + \eta$. Then, by applying a first-order Taylor expansion of f in a local neighborhood of i_{high}, we have:

$$x' = f\left(i_{high} + \Delta\right) \approx f\left(i_{high}\right) + J_f\left(i_{high}\right)\Delta, \tag{53}$$

$$x' \approx x + J_f\left(i_{high}\right)(A - I)\, i_{high} + J_f\left(i_{high}\right)\eta. \tag{54}$$

Accordingly, assuming that the source and degraded features retain local correspondence, the induced source-to-target transformation in feature space can be locally linearized as:

$$x' \approx A_\Omega\, x + b_\Omega + \varepsilon_\Omega, \tag{55}$$

where A_Ω denotes the Jacobian of the locally induced feature-space transformation, b_Ω is the corresponding local offset, and ε_Ω collects the propagated noise term $J_f\left(i_{\text{high}}\right)\eta$ and higher-order Taylor residuals.

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3) *Cross-weather perturbation:* Under weather conditions such as haze, cloud, rain, and snow, features within a local window Ω can be characterized by contrast compression, neighborhood scattering mixing, and anisotropic noise:

$$x' \approx \alpha_\Omega x + \beta_\Omega \mathbf{1} + \sum_{k \in N} \gamma_k x_k + \varepsilon_\Omega, \tag{56}$$

where $\varepsilon_\Omega \sim \mathcal{N}(0, \Sigma_{\varepsilon,\Omega})$, $\Sigma_{\varepsilon,\Omega}$ characterizes direction-dependent anisotropic noise, $\alpha_\Omega \in (0,1)$ denotes dynamic-range compression, $\beta_\Omega \mathbf{1}$ represents the bias uplift induced by scattering, and $\sum_{k \in N} \gamma_k x_k$ describes neighborhood background mixing. After window-level vectorization, Equation (56) takes the LAP form in Equation (1), where A_Ω characterizes dynamic-range compression and neighborhood mixing, and b_Ω represents the scattering-induced bias.

APPENDIX B
ATTRIBUTE PERTURBATION DETAILS OF TDKV

For reproducibility, this appendix provides the domain-attribute sets, sampling distributions, and noise operations used in TDKV. Let p denote a clean attribute-aware prompt containing the object category c and domain attributes drawn from the domain-attribute set $\mathcal{D}$ defined below. TDKV preserves c and generates noisy textual key-value tokens by perturbing only the attribute phrases in p.

The perturbation operator $\mathcal{C}(\cdot)$ used in the main text is specified as:

$$\mathcal{C}(p; \mathcal{D}) = \left(T^{\xi}_{\text{contr}} \circ T^{\rho}_{\text{shuffle}} \circ T^{\gamma}_{\text{drop}} \right)(p; \mathcal{D}),$$

where T^{γ}_{drop} deletes an attribute phrase, $T^{\rho}_{\text{shuffle}}$ reorders the attribute phrases, and T^{ξ}_{contr} replaces an attribute with another value from the same attribute group. The domain-attribute set is defined as:

$$\begin{aligned}
\mathcal{D} &= \{\mathcal{D}_{\text{sce}}, \mathcal{D}_{\text{wea}}, \mathcal{D}_{\text{plat}}, \mathcal{D}_{\text{ill}}\}, \\
\mathcal{D}_{\text{sce}} &= \{\text{airport}, \text{transportation}, \text{port}, \\
&\qquad \text{natural area}, \ \text{residential area}\}, \\
\mathcal{D}_{\text{wea}} &= \{\text{sunny}, \text{rainy}, \text{snowy}, \text{hazy}\}, \\
\mathcal{D}_{\text{plat}} &= \{\text{satellite}, \text{drone}, \text{camera}\}, \\
\mathcal{D}_{\text{ill}} &= \{\text{daytime}, \text{night}\}.
\end{aligned}$$

The probabilities of the three noise operations are set as:

$$\gamma = 0.15, \qquad \rho = 0.10, \qquad \xi = 0.10.$$

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